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Subject: DevOps
Case Study : 86

EduGov – National Digital Education Infrastructure

1. Problem Statement:

EduGov Digital Services manages a nationwide education platform that supports online learning, digital examinations, certification management, admissions, academic records, scholarship processing, and research collaboration services.

Due to increasing user traffic during examination periods, admission cycles, and national assessments, the existing infrastructure faced challenges including system outages, delayed processing, limited scalability, inconsistent deployment practices, and security risks.

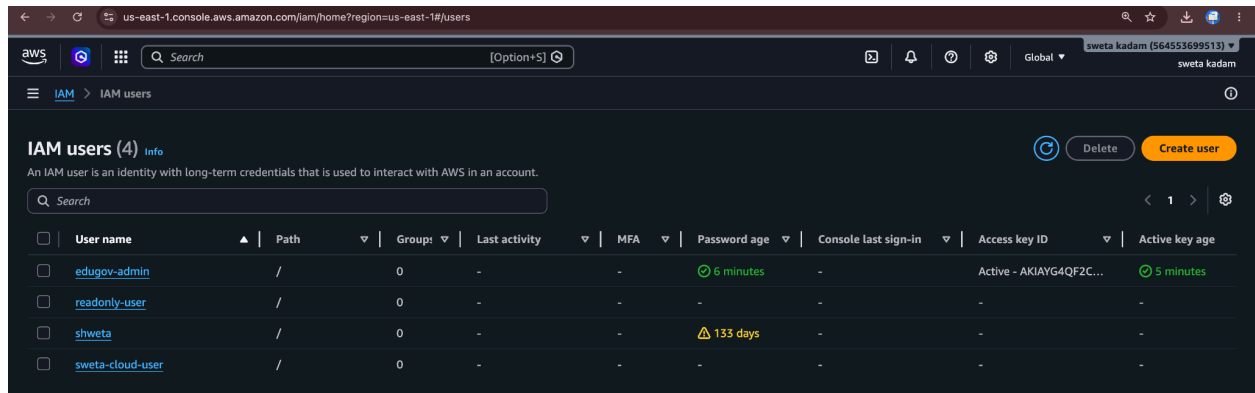
The objective of this project was to design and implement a cloud-native DevOps ecosystem capable of supporting large-scale educational services with high availability, automation, security, monitoring, centralized logging, disaster recovery, and continuous delivery using modern DevOps tools and practices.

2. Project Objectives:

- Automate infrastructure provisioning using Terraform
- Containerize applications using Docker
- Deploy applications on Kubernetes
- Implement CI/CD using Jenkins
- Configure monitoring using Prometheus and Grafana
- Implement centralized logging using ELK Stack
- Manage secrets using HashiCorp Vault
- Ensure scalability, security, and disaster recovery

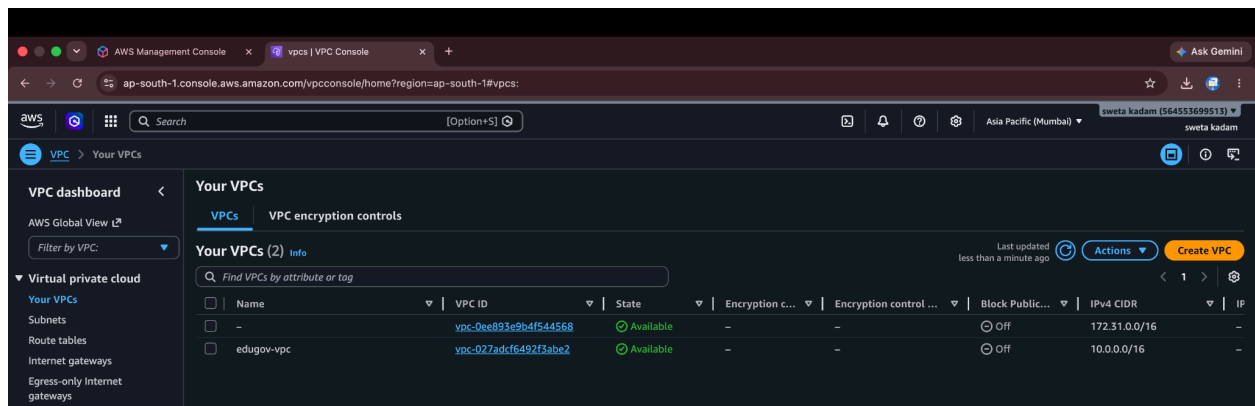
3. Project Implementation:

3.1 IAM User Creation for EduGov Infrastructure



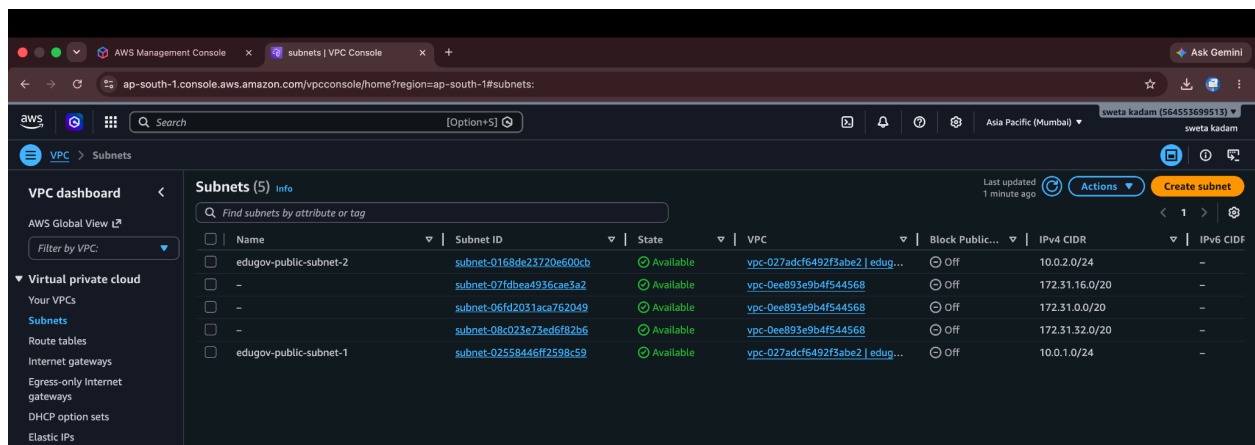
Created IAM user edugov-admin with programmatic access to securely manage AWS resources through the AWS CLI and console.

3.2 Virtual Private Cloud (VPC) Setup



Created custom VPC edugov-vpc (10.0.0.0/16) to provide an isolated and secure network environment for the EduGov infrastructure.

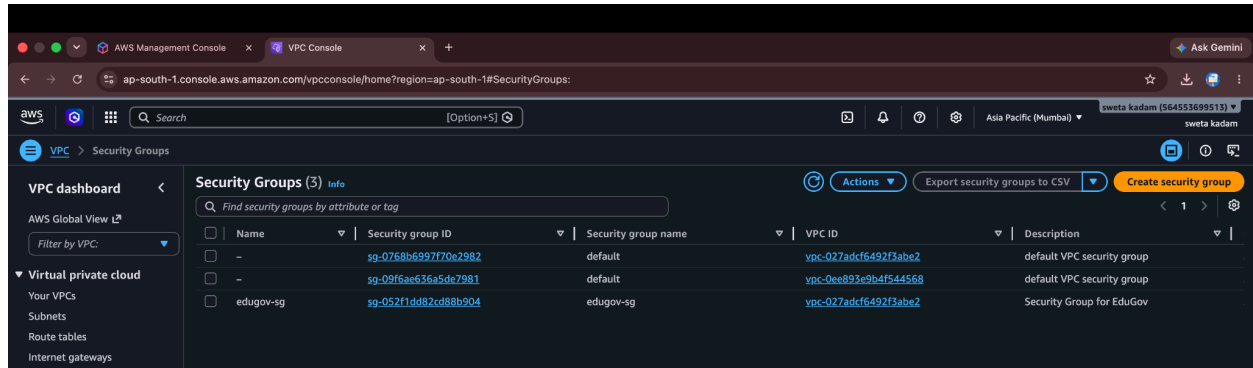
3.3 Configuration of Public and Private Subnets for the EduGov VPC



Public and private subnets were configured within the EduGov VPC to implement a secure network architecture. Public subnets host internet-facing resources, while private

subnets are reserved for internal services and backend workloads, enhancing security and network isolation.

3.4 Network Security Configuration



Created a security group named `edugov-sg` to define and enforce network access policies. The security group controls inbound and outbound traffic for EC2 instances and application services deployed within the EduGov environment.

3.5 AWS CLI Configuration and IAM User Verification

```
/opt/homebrew/share/awscli/examples
[swetapopatkadam@swetas-MacBook-Air ~ % aws --version
aws-cli/2.35.1 Python/3.14.5 Darwin/25.3.0 source/arm64
[swetapopatkadam@swetas-MacBook-Air ~ % aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

[AWS Access Key ID [None]: AKIAYG4QF2C42QA6TNWV
[AWS Secret Access Key [None]: wFWgQ0nq59LDDzQ2Rmxqd2IHBh5Mt8y9Cqo3NF0+
Default region name [None]: ap-south-1
Default output format [None]: json
[swetapopatkadam@swetas-MacBook-Air ~ % aws sts get-caller-identity
{
  "UserId": "AIDAYG4QF2C4ZKQYIXIQV",
  "Account": "564553699513",
  "Arn": "arn:aws:iam::564553699513:user/edugov-admin"
}
[swetapopatkadam@swetas-MacBook-Air ~ % ]
```

Configured the AWS Command Line Interface (AWS CLI) using the access credentials of the `edugov-admin` IAM user. The configuration included setting the default AWS region (`ap-south-1`) and output format (JSON). User authentication and permissions were successfully verified using the `aws sts get-caller-identity` command, confirming secure access to the EduGov AWS account through the AWS CLI.

3.6 Terraform Initialization

Created the Terraform project structure and initialized the Terraform working directory using `terraform init`. The AWS provider plugin was successfully downloaded and configured, preparing the environment for Infrastructure as Code (IaC) deployment.

```

swetapopatkadam@swetas-MacBook-Air ~ % cd ~/Desktop/devops/EduGov
swetapopatkadam@swetas-MacBook-Air EduGov % mkdir terraform
swetapopatkadam@swetas-MacBook-Air EduGov % cd terraform
swetapopatkadam@swetas-MacBook-Air terraform % touch provider.tf
touch main.tf
touch variables.tf
touch outputs.tf
swetapopatkadam@swetas-MacBook-Air terraform % ls
main.tf      outputs.tf   provider.tf   variables.tf
swetapopatkadam@swetas-MacBook-Air terraform % nano provider.tf
swetapopatkadam@swetas-MacBook-Air terraform % aws sts get-caller-identity
{
  "UserId": "AIDAY64QF2C4ZKQYIXIQV",
  "Account": "564553699513",
  "Arn": "arn:aws:iam::564553699513:user/edugov-admin"
}
swetapopatkadam@swetas-MacBook-Air terraform % terraform init
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.49.0...
- Installed hashicorp/aws v6.49.0 (signed by HashiCorp)

Initializing the backend...

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
swetapopatkadam@swetas-MacBook-Air terraform % █

```

3.7 Terraform Validation and Infrastructure Planning

```

commands will detect it and remind you to do so if necessary.
swetapopatkadam@swetas-MacBook-Air terraform % nano main.tf
swetapopatkadam@swetas-MacBook-Air terraform % terraform validate
Success! The configuration is valid.

swetapopatkadam@swetas-MacBook-Air terraform % terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
+ create

Terraform will perform the following actions:

# aws_internet_gateway.igw will be created
+ resource "aws_internet_gateway" "igw" {
+   arn      = (known after apply)
+   id       = (known after apply)
+   owner_id = (known after apply)
+   region   = "ap-south-1"
+ }

```

Validated the Terraform configuration using `terraform validate` and generated an execution plan using `terraform plan`. The plan confirmed that the required AWS resources, including the VPC, subnets, and Internet Gateway, would be created successfully.

3.8 Infrastructure Provisioning Using Terraform

The Terraform configuration using `terraform apply`, resulting in the successful creation of the EduGov VPC, public subnets, and Internet Gateway. Terraform reported successful resource provisioning with no errors.

3.9 Secure SSH Access to EC2 Instance

Established a secure SSH connection to the Ubuntu EC2 instance using the generated PEM key. This verified successful network configuration, security group rules, and remote server accessibility.

```
ssh: connect to host 65.2.35.188 port 22: Operation timed out
swetapopatkadam@swetas-MacBook-Air Downloads % ssh -i ~/Downloads/edugov-key.pem ubuntu@65.2.35.188
The authenticity of host '65.2.35.188 (65.2.35.188)' can't be established.
ED25519 key fingerprint is: SHA256:UnYQ5/9/HpwzteAnZc05aXDe3LTQZ9rDI1WZb18xNxx
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '65.2.35.188' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jun 10 20:16:13 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.4% of 6.61GB Processes:    115
Memory usage: 26%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.2.161

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-2-161:~$
```

3.10 EC2 Server Preparation and Docker Installation

```
Get:47 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [33.1 kB]
Get:48 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [42.7 kB]
Get:49 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [2860 B]
Get:50 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [201 kB]
Get:51 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [35.1 kB]
Get:52 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 c-n-f Metadata [396 B]
Get:53 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:54 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Fetched 32.5 MB in 4s (8067 kB/s)
47 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-10-0-2-161:~$ sudo apt install docker.io -y
Installing:
  docker.io

Installing dependencies:
  bridge-utils  containerd  dns-root-data  dnsmasq-base  pigz  runc  ubuntu-fan

Suggested packages:
  ifupdown  cgroupfs-mount  debootstrap  docker-compose-v2  rinse  zfs-fuse
  aufs-tools  | cgroup-lite  docker-buildx  docker-doc  rootlesskit  | zfsutils

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 47
  Download size: 74.0 MB
  Space needed: 201 MB / 467 MB available
```

Updated the Ubuntu package repository and installed Docker Engine on the EC2

instance. Docker service was started, enabled for automatic startup, and verified to ensure the server was ready for containerized application deployment.

3.11 Source Code Deployment from GitHub Repository

Installed Git and cloned the EduGov DevOps project repository from GitHub onto the EC2 instance. This made the application source code available on the server for further build and deployment activities.

```
Setting up ubuntu-fan (0.12.17) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/ubuntu-fan.service' → '/usr/lib/systemd/system/ubuntu-fan.service'.
Setting up docker.io (29.1.3-0ubuntu4.1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' → '/usr/lib/systemd/system/docker.service'.
Created symlink '/etc/systemd/system/sockets.target.wants/docker.socket' → '/usr/lib/systemd/system/docker.socket'.
Processing triggers for dbus (1.16.2-2ubuntu4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

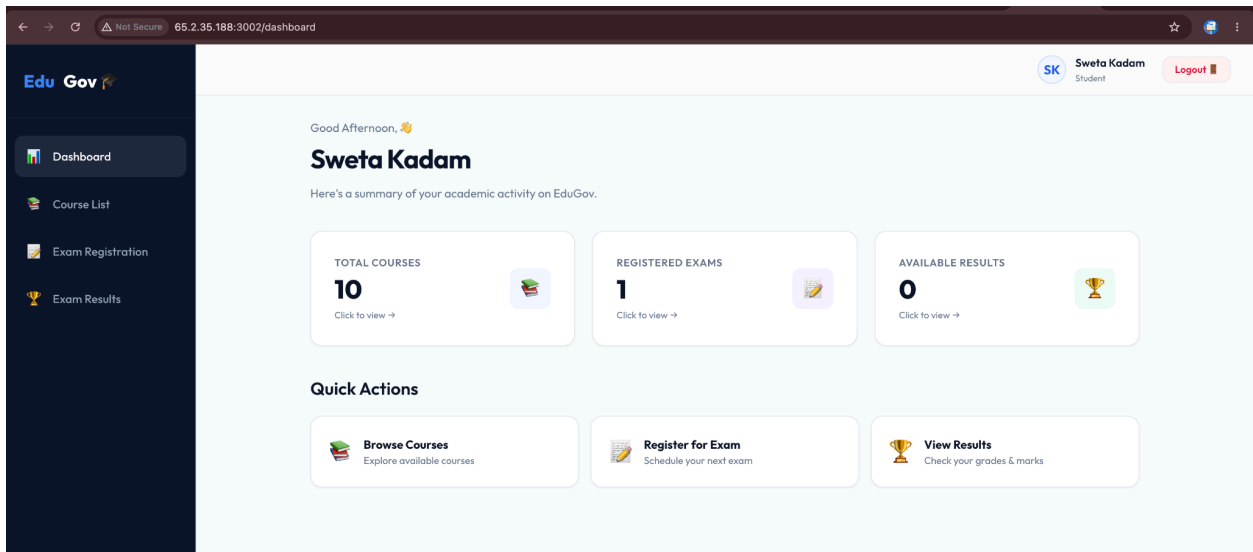
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-2-161:~$ sudo systemctl start docker
ubuntu@ip-10-0-2-161:~$ sudo systemctl enable docker
ubuntu@ip-10-0-2-161:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-10-0-2-161:~$ sudo apt install git -y
git clone https://github.com/Shweta-Tech-creator/EduGov-DevOps.git
cd EduGov-DevOps
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 47
Cloning into 'EduGov-DevOps'...
remote: Enumerating objects: 68, done.
remote: Counting objects: 100% (68/68), done.
remote: Compressing objects: 100% (54/54), done.
remote: Total 68 (delta 8), reused 68 (delta 8), pack-reused 0 (from 0)
Receiving objects: 100% (68/68), 140.75 KiB | 8.28 MiB/s, done.
Resolving deltas: 100% (8/8), done.
ubuntu@ip-10-0-2-161:~/EduGov-DevOps$ ls
README.md  backend  database  frontend  package-lock.json
ubuntu@ip-10-0-2-161:~/EduGov-DevOps$
```

3.12 Application Deployment Verification

Verified successful deployment of the EduGov backend application by accessing the public EC2 endpoint. The API returned a successful response, confirming that the application was running correctly and accessible over the network.

```
← → ↻ ⚠ Not Secure 65.2.35.188:5001
Pretty print ☒
{
  "success": true,
  "message": "Welcome to the EduGov Education Management System API"
}
```



3.13 EduGov Application Container Deployment and Verification

```
ubuntu@ip-10-0-2-161:~/EduGov-DevOps/frontend$ sudo docker images
NAME                ID                                IMAGE                                DISK USAGE    CONTENT SIZE    EXTRA
edugov-backend:latest e4f43ce29221                    1.6GB          402MB          U
edugov-frontend:latest 47679e1065cf                    1.67GB         422MB          U
mongo:7               c1a84ab5d0c1                    1.19GB         296MB          U
node:18                c6ae79e38498                    1.58GB         411MB          U

ubuntu@ip-10-0-2-161:~/EduGov-DevOps/frontend$ sudo docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
a7c91b5c1702   edugov-frontend                    "docker-entrypoint.s..." 2 minutes ago  Up 2 minutes  0.0.0.0:3002->3000/tcp, [::]:3002->3000/tcp
67bebd9ebaf1   edugov-backend                    "docker-entrypoint.s..." 11 hours ago   Exited (1) 15 minutes ago
81db935b591c   mongo:7                             "docker-entrypoint.s..." 11 hours ago   Exited (0) 15 minutes ago

ubuntu@ip-10-0-2-161:~/EduGov-DevOps/frontend$ sudo docker start mongodb
mongodb

ubuntu@ip-10-0-2-161:~/EduGov-DevOps/frontend$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
a7c91b5c1702   edugov-frontend                    "docker-entrypoint.s..." 3 minutes ago  Up 3 minutes  0.0.0.0:3002->3000/tcp, [::]:3002->3000/tcp
81db935b591c   mongo:7                             "docker-entrypoint.s..." 11 hours ago   Up 6 seconds  0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp

ubuntu@ip-10-0-2-161:~/EduGov-DevOps/frontend$ sudo docker start backend
backend

ubuntu@ip-10-0-2-161:~/EduGov-DevOps/frontend$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
a7c91b5c1702   edugov-frontend                    "docker-entrypoint.s..." 3 minutes ago  Up 3 minutes  0.0.0.0:3002->3000/tcp, [::]:3002->3000/tcp
67bebd9ebaf1   edugov-backend                    "docker-entrypoint.s..." 11 hours ago   Up 12 seconds  0.0.0.0:5001->5000/tcp, [::]:5001->5000/tcp
81db935b591c   mongo:7                             "docker-entrypoint.s..." 11 hours ago   Up 27 seconds  0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp

ubuntu@ip-10-0-2-161:~/EduGov-DevOps/frontend$
```

The EduGov application services were successfully deployed using Docker containers. The frontend, backend, and MongoDB containers were started and verified using Docker commands. Container status and port mappings confirmed that all services were running correctly and ready for application access.

3.14 Kubernetes Cluster Deployment Verification

Verified the successful deployment of EduGov on the K3s Kubernetes cluster. The output shows running pods, active services, and a healthy cluster node, confirming that the application infrastructure is operational.

```
ubuntu@ip-10-0-1-43:~$ sudo k3s kubectl get pods -n default
NAME                                READY   STATUS    RESTARTS   AGE
edugov-backend-75d58cbfd4-jvkzp    0/1     Running   14 (12m ago)   3h9m
edugov-frontend-58647bc8df-7mmnk   0/1     Running   3 (6m56s ago)  24m
edugov-frontend-58647bc8df-9th4b   0/1     Running   3 (4m38s ago)  24m
edugov-frontend-58647bc8df-9zjm4   0/1     Running   2 (12m ago)    24m
edugov-frontend-58647bc8df-c6tvw   1/1     Running   9 (12m ago)    3h9m
mongodb-7c8876d98b-j4rmz           1/1     Running   1 (51m ago)    3h9m
ubuntu@ip-10-0-1-43:~$ sudo k3s kubectl get svc -n default
NAME                TYPE            CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
edugov-backend      LoadBalancer   10.43.173.230   10.0.1.43        5001:31918/TCP   3h10m
edugov-frontend     LoadBalancer   10.43.174.163   10.0.1.43        3000:30224/TCP   3h10m
elasticsearch        ClusterIP       10.43.105.184   <none>           9200/TCP         175m
kibana               NodePort       10.43.197.188   <none>           5601:31000/TCP   175m
kubernetes           ClusterIP       10.43.0.1       <none>           443/TCP          3h19m
mongodb              ClusterIP       10.43.167.31    <none>           27017/TCP        3h10m
ubuntu@ip-10-0-1-43:~$
```

3.15 Kubernetes Pods and Services Status

Verified the status of deployed EduGov pods and services using Kubernetes commands. The output confirms that application components are running and accessible through configured services.

```
ubuntu@ip-10-0-1-43:~$ sudo k3s kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
ip-10-0-1-43        Ready     control-plane   175m   v1.35.5+k3s1
```

3.16 Elasticsearch & Kibana Deployment Verification

Elasticsearch and Kibana were successfully deployed in the K3s cluster, with both deployments showing the desired replica count and running status.

```
ubuntu@ip-10-0-1-43:~/EduGov-DevOps$ sudo k3s kubectl get deployments -n default | grep -E 'elasticsearch|kibana'
ubuntu@ip-10-0-1-43:~/EduGov-DevOps$ sudo k3s kubectl apply -f logging/elasticsearch.yaml
[sudo k3s kubectl apply -f logging/kibana.yaml]
deployment.apps/elasticsearch created
service/elasticsearch created
deployment.apps/kibana created
service/kibana created
ubuntu@ip-10-0-1-43:~/EduGov-DevOps$ sudo k3s kubectl get deployments -n default | grep -E 'elasticsearch|kibana'
elasticsearch      1/1      1          15s
kibana             1/1      1          15s
ubuntu@ip-10-0-1-43:~/EduGov-DevOps$
```

3.17 Prometheus & Grafana Monitoring Setup

The kube-prometheus-stack was deployed successfully, and all monitoring components, including Prometheus, Grafana, and Node Exporter, were running in the monitoring namespace.

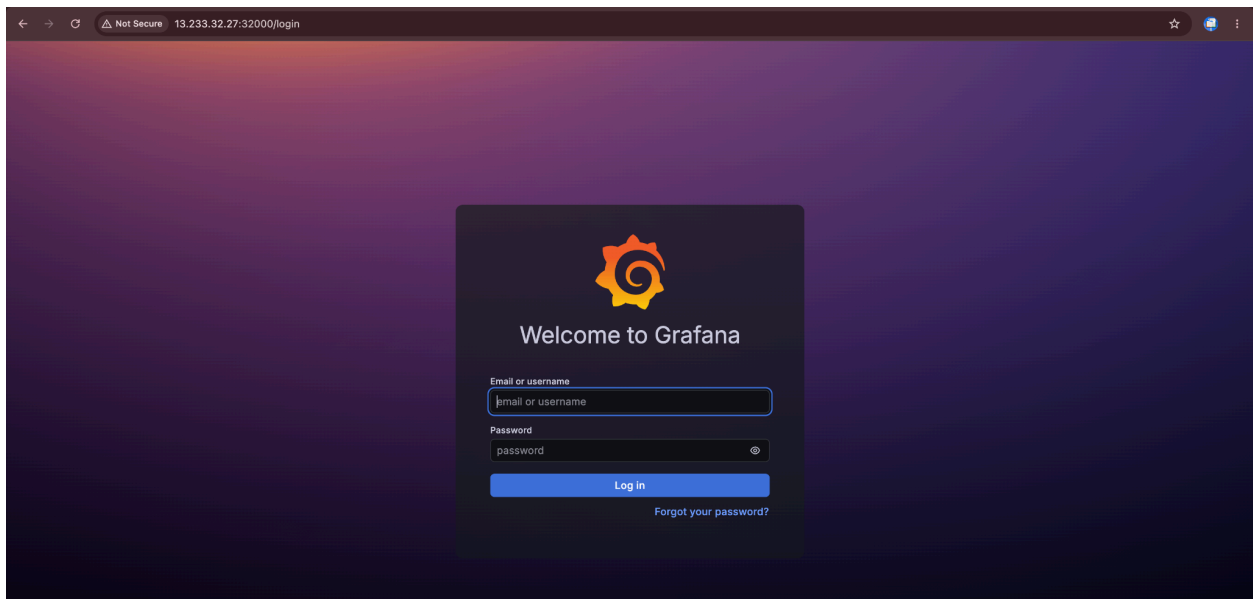
```

ubuntu@ip-10-0-1-43:~/EduGov-DevOps/monitoring$ sudo k3s kubectl get pods -n monitoring
NAME                                READY   STATUS    RESTARTS   AGE
kube-prometheus-stack-grafana-75b5649fc4-9s9nr    2/3     Running   0           15s
kube-prometheus-stack-kube-state-metrics-58f5d9c5d4-z8lzj  1/1     Running   0           15s
kube-prometheus-stack-operator-79fd575db7-9pcns    1/1     Running   0           15s
kube-prometheus-stack-prometheus-node-exporter-gqslx  1/1     Running   0           15s
prometheus-kube-prometheus-stack-prometheus-0     0/2     PodInitializing   0           8s
ubuntu@ip-10-0-1-43:~/EduGov-DevOps/monitoring$ sudo k3s kubectl get pods -n monitoring
NAME                                READY   STATUS    RESTARTS   AGE
kube-prometheus-stack-grafana-75b5649fc4-9s9nr    2/3     Running   0           62s
kube-prometheus-stack-kube-state-metrics-58f5d9c5d4-z8lzj  1/1     Running   0           62s
kube-prometheus-stack-operator-79fd575db7-9pcns    1/1     Running   0           62s
kube-prometheus-stack-prometheus-node-exporter-gqslx  1/1     Running   0           62s
prometheus-kube-prometheus-stack-prometheus-0     2/2     Running   0           55s
ubuntu@ip-10-0-1-43:~/EduGov-DevOps/monitoring$ helm list -n monitoring
NAME                APP VERSION   NAMESPACE   REVISION   UPDATED                               STATUS   CHART
kube-prometheus-stack 86.2.2        monitoring   1          2026-06-13 19:40:15.103096515 +0000 UTC deployed kube-prometheus-st
ack-86.2.2          v0.91.0
ubuntu@ip-10-0-1-43:~/EduGov-DevOps/monitoring$

```

3.18 Grafana Dashboard Access

Grafana was accessed through the browser, confirming successful exposure of the monitoring dashboard and readiness for metrics visualization.



3.19 Elasticsearch Cluster Health Verification

Verified the Elasticsearch deployment by accessing the cluster health endpoint. The cluster status was green, with all primary and active shards available, confirming successful deployment and healthy operation of the logging infrastructure.

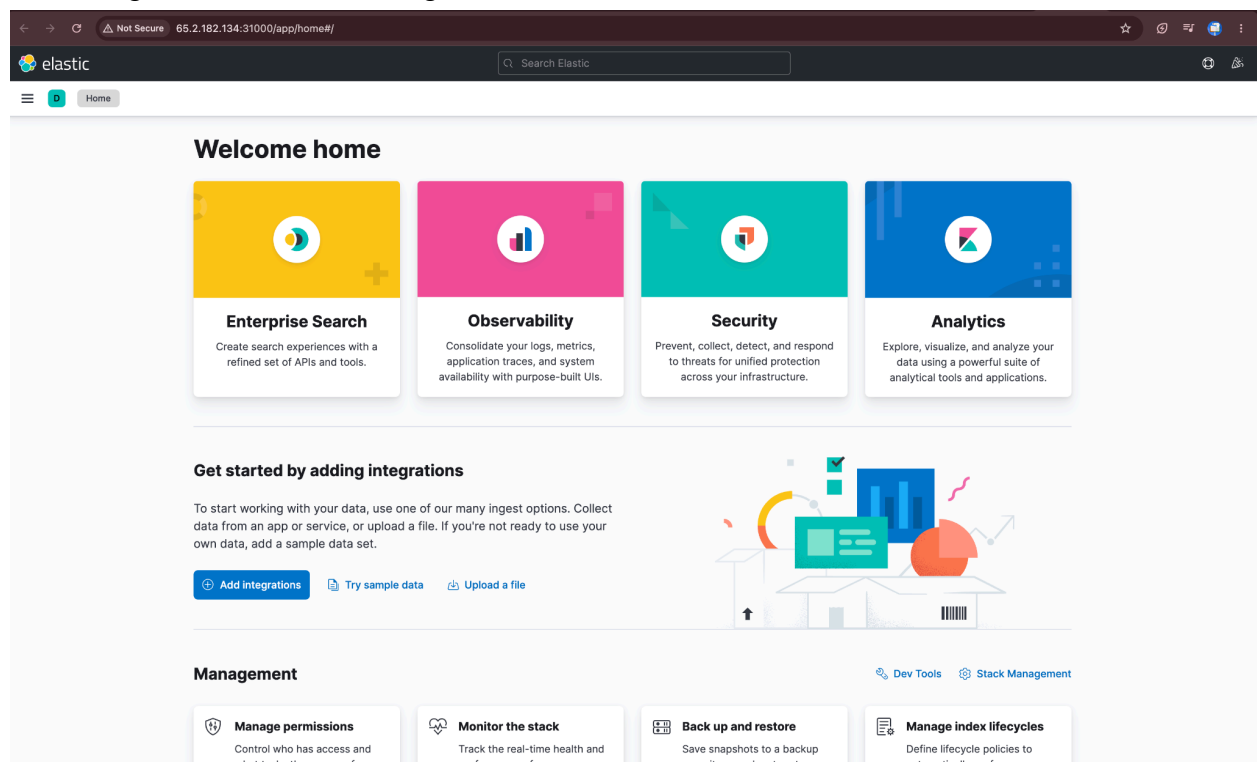
```
← → ↻ ⚠ Not Secure 65.2.182.134:30920/_cluster/health?pretty

Pretty print ☒

{
  "cluster_name": "docker-cluster",
  "status": "green",
  "timed_out": false,
  "number_of_nodes": 1,
  "number_of_data_nodes": 1,
  "active_primary_shards": 9,
  "active_shards": 9,
  "relocating_shards": 0,
  "initializing_shards": 0,
  "unassigned_shards": 0,
  "delayed_unassigned_shards": 0,
  "number_of_pending_tasks": 0,
  "number_of_in_flight_fetch": 0,
  "task_max_waiting_in_queue_millis": 0,
  "active_shards_percent_as_number": 100
}
```

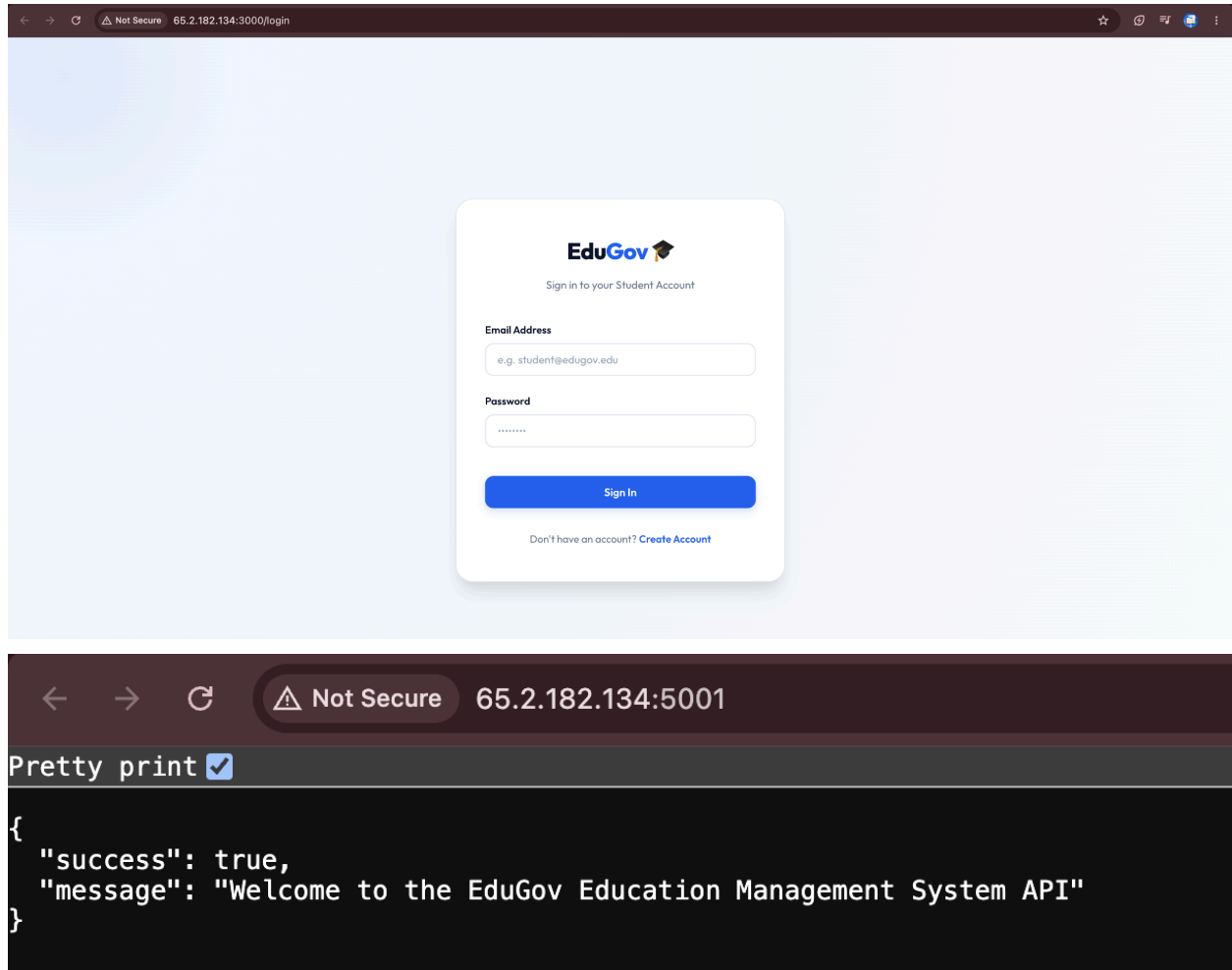
3.20 Kibana Dashboard Access Verification

Successfully accessed the Kibana web interface, confirming that the Kibana deployment is running and available for log visualization.



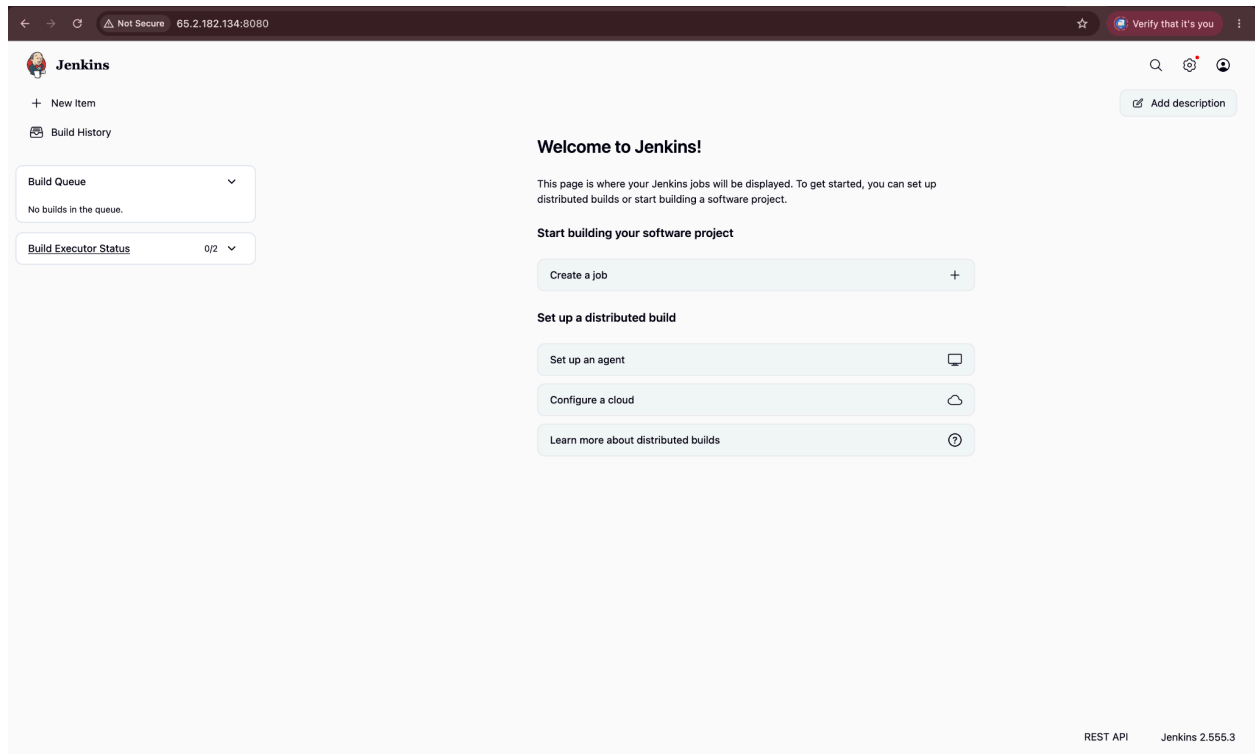
3.21 Ingress Configuration for Frontend and Backend Services

Configured and deployed Kubernetes Ingress to provide external access to both frontend and backend services through a unified routing layer. Verified successful access to application endpoints via the Ingress controller.



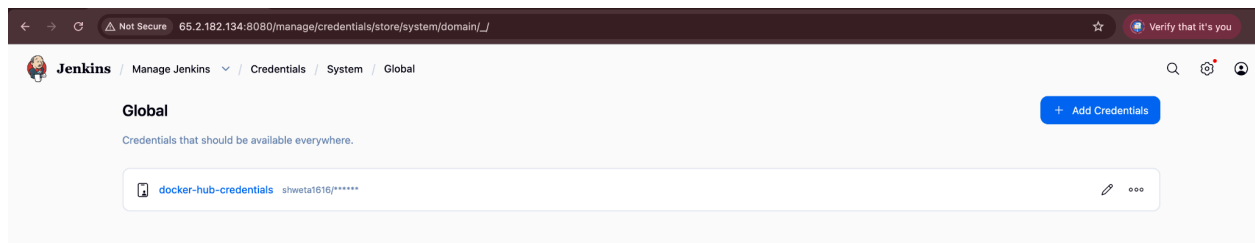
3.22 Jenkins CI/CD Pipeline Deployment

Successfully deployed and accessed the Jenkins web interface, confirming that the CI/CD automation server is operational and ready for pipeline management.



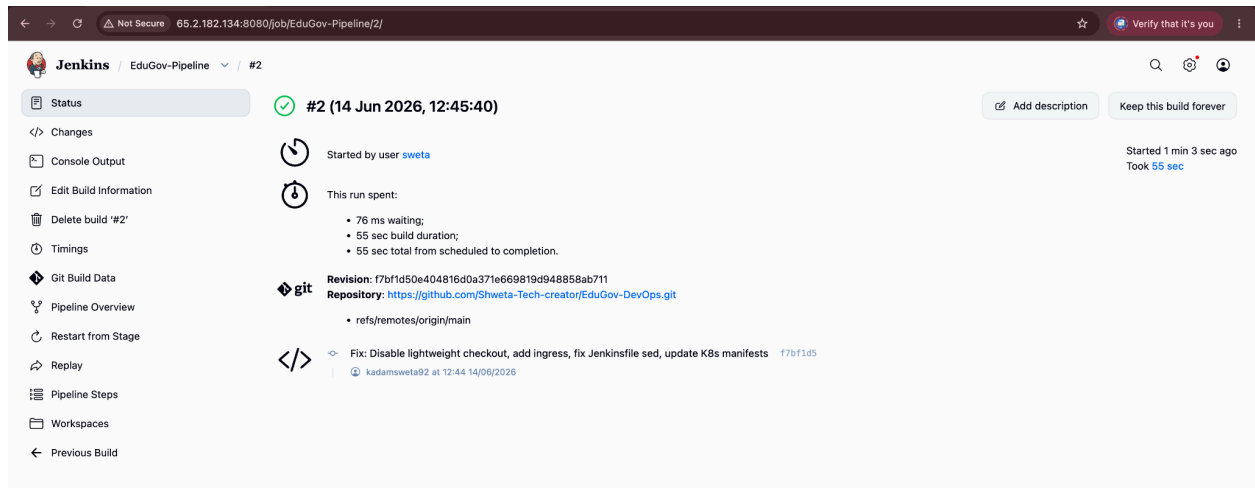
3.23 Jenkins Global Credentials Configuration

Configured global credentials in Jenkins to enable secure authentication with external services and repositories required for automated CI/CD pipeline execution.



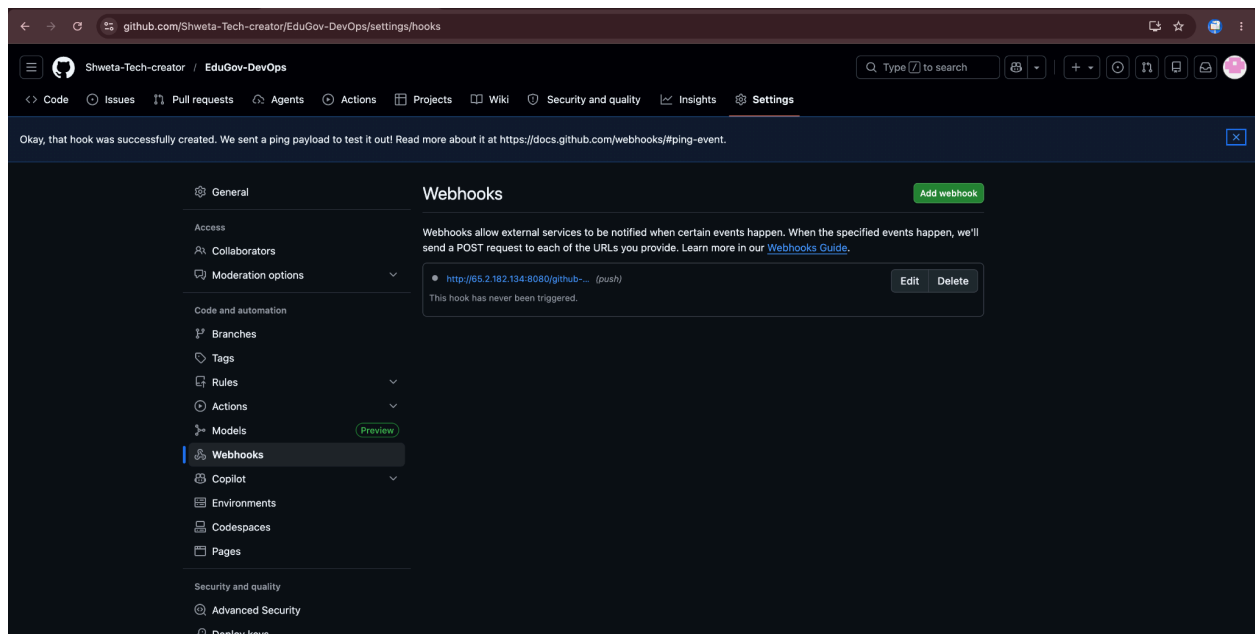
3.24 Jenkins Pipeline Creation and Configuration

Created and configured a Jenkins pipeline to automate the build, deployment, and management of the EduGov application. The pipeline integrates source code retrieval, container deployment, and continuous delivery processes.



3.25 GitHub Webhook Configuration for Jenkins Integration

Configured GitHub webhooks to automatically trigger Jenkins pipeline executions whenever code changes are pushed to the repository. This enables continuous integration and automated deployment of the EduGov application.



3.26 Webhook-Triggered Jenkins Pipeline Execution

Verified the integration between GitHub and Jenkins through webhooks. Upon pushing code changes to the repository, Jenkins automatically triggered the pipeline, demonstrating successful Continuous Integration (CI) automation.

Jenkins

EduGov-Pipeline

#3

Status

Changes

Console Output

Edit Build Information

Delete build '#3'

Polling Log

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

#3 (14 Jun 2026, 12:51:33)

Started by GitHub push by Shweta-Tech-creator

This run spent:

- 6.1 sec waiting;
- 43 sec build duration;
- 49 sec total from scheduled to completion.

Revision: 37f4301e6309bb1e4aab7dcd30469009e173f2c8

Repository: <https://github.com/Shweta-Tech-creator/EduGov-DevOps.git>

- refs/remotes/origin/main

Cl: Trigger Jenkins pipeline build - test webhook integration 37f4381

kadamsweta92 at 12:51 14/06/2026

Add description

Keep this build forever

Started 53 sec ago

Took 43 sec

3.27 HashiCorp Vault Secret Injection Verification

Verified successful secret injection from HashiCorp Vault into the EduGov backend application, ensuring secure access to sensitive configuration data.

```

ubuntu@ip-10-0-1-43:~$ sudo kubectl get pods | grep vault
edugov-backend-vault-7bdb87cfb5-cq289    2/2    Running    0          5m35s
vault-agent-injector-7c8fbb8684-mt2df    1/1    Running    0          15m
ubuntu@ip-10-0-1-43:~$ sudo kubectl exec deployment/edugov-backend-vault -c edugov-backend -- cat /vault/
secrets/db-creds
MONGO_URI="mongodb://mongodb:27017/edugov_db"
JWT_SECRET="edugov_super_secret_jwt_key_2026"
ubuntu@ip-10-0-1-43:~$ sudo kubectl logs deployment/edugov-backend-vault -c edugov-backend --tail=15
Found 6 pods, using pod/edugov-backend-vault-7bdb87cfb5-cq289

> edugov-backend@1.0.0 start
> node index.js

=====
EduGov Backend Server started successfully!
Local Access: http://localhost:5000
=====
MongoDB Connected: mongodb -> Database: edugov_db
ubuntu@ip-10-0-1-43:~$

```

3.28 Horizontal Pod Autoscaler (HPA) Verification

Verified Kubernetes Horizontal Pod Autoscaler (HPA) configuration and observed multiple running backend pods, demonstrating the scalability and high-availability capabilities of the EduGov application

```

ubuntu@ip-10-0-1-43:~$ sudo kubectl get hpa
NAME                REFERENCE                TARGETS          MINPODS   MAXPODS   REPLICAS   AGE
edugov-backend-hpa  Deployment/edugov-backend  cpu: 3%/50%      1          5          5           70m
ubuntu@ip-10-0-1-43:~$ sudo kubectl get pods -l app=edugov-backend
NAME                                READY   STATUS    RESTARTS   AGE
edugov-backend-6498c7cf87-617wc     1/1     Running   0           3m25s
edugov-backend-6498c7cf87-764wm     1/1     Running   0           3m25s
edugov-backend-6498c7cf87-9v745     1/1     Running   0           6m14s
edugov-backend-6498c7cf87-ckbkr     1/1     Running   0           3m11s
edugov-backend-6498c7cf87-xr6l8     1/1     Running   0           3m25s
edugov-backend-vault-7bdb87cfb5-cq289 2/2     Running   0           6m14s
ubuntu@ip-10-0-1-43:~$

```


4. Conclusion

The EduGov – National Digital Education Infrastructure project was successfully designed and implemented using modern DevOps practices and cloud-native technologies. Infrastructure provisioning was automated using Terraform, application services were containerized using Docker, and deployments were orchestrated through Kubernetes (K3s). Continuous Integration and Continuous Deployment (CI/CD) were achieved using Jenkins with GitHub Webhook integration.

To ensure operational visibility and reliability, Prometheus and Grafana were implemented for monitoring, while Elasticsearch and Kibana provided centralized logging and log analysis capabilities. HashiCorp Vault was integrated for secure secret management, eliminating the need to store sensitive credentials within application code. Additionally, Horizontal Pod Autoscaling (HPA) was configured to support application scalability and high availability under varying workloads.

The final solution delivers a secure, automated, scalable, and resilient digital education platform capable of supporting large-scale educational services while meeting the objectives defined in the project requirements.

Repository Link:

<https://github.com/Shweta-Tech-creator/EduGov-DevOps>